

Introduction to Earth System

Feedback processes & the Daisy world

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1

1

I. Feedback

Definition

- A process where an initial change in the climate triggers a secondary change that either **reinforces** or **dampens** the original change

Positive feedback

Negative feedback

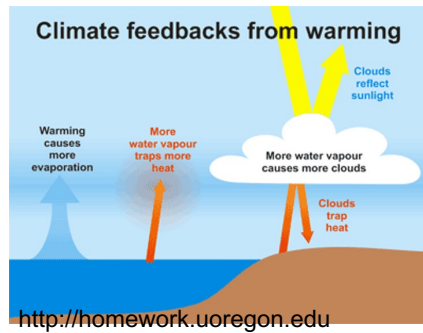
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2

2

Water Vapor Feedback

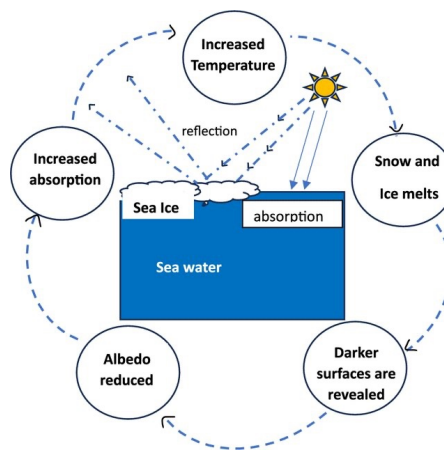
- surface temperatures $\uparrow \rightarrow$ evaporation $\uparrow \rightarrow$ more water vapor (warmer air also holds more water vapor) $\rightarrow$ traps more heat $\rightarrow$ further increases temperatures



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3

Ice-albedo feedback



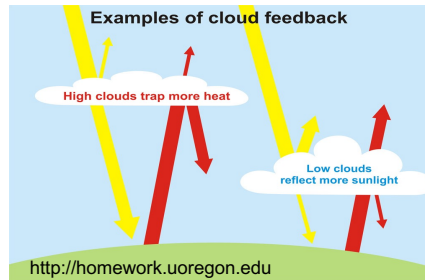
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4

4

Cloud-albedo feedback

- **Low cloud:**
 - Temperature $\uparrow \rightarrow$ evaporation $\uparrow \rightarrow$ cloud cover $\uparrow \rightarrow$ less solar energy available to heat the atmosphere $\rightarrow$ temperature $\downarrow$
- **High cloud?**



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5

Lapse-rate feedback

- **Physical basis:** The lapse rate is the rate at which atmospheric temperature decreases with increasing altitude in the troposphere.

$$\Gamma = -dT/dz$$

- **Lapse rate feedback:** occurs when the warming of the atmosphere is vertically non-uniform:
 - If higher levels warm more than the surface $\rightarrow$ **negative feedback** (dampening).
 - If the surface warms more than higher levels $\rightarrow$ **positive feedback** (amplifying).

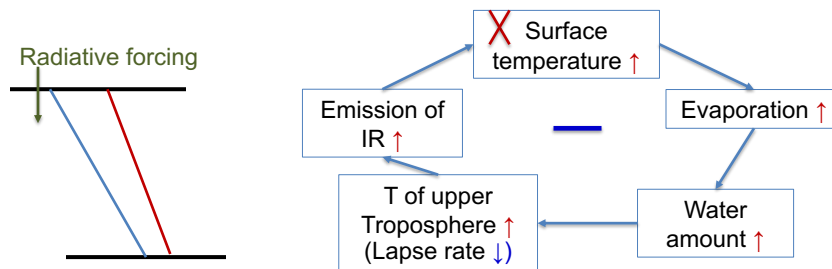
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6

6

Lapse-rate feedback

- **Negative feedback** loop: Tropical lapse rate
 - ✓ **Surface Warming** → Increases evaporation & deep convection.
 - ✓ **Latent Heat Release** → Warm, moist air rises and releases heat at high altitudes.
 - ✓ **Upper Troposphere Warming** → Warms faster than the surface
 - ✓ **Enhanced Cooling**: Warmer upper atmosphere radiates heat to space more efficiently.
 - **Result**: slows down the initial surface warming.



- The positive water vapor feedback is partially offset by the globally negative lapse rate feedback

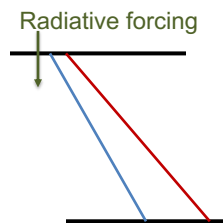
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7

7

Lapse-rate feedback

- **Positive feedback** loop: polar lapse rate
 - ✓ **Stable Stratification**: Polar atmosphere traps surface warming near the ground → Heat does not easily reach the upper troposphere.
 - ✓ **Uneven Warming**: The surface warms significantly more than the upper layers.
 - ✓ **Reduced Cooling**: Cool upper atmosphere fails to radiate extra heat to space efficiently.
 - **Result**: Amplifies surface warming → a key driver of **Arctic Amplification**.

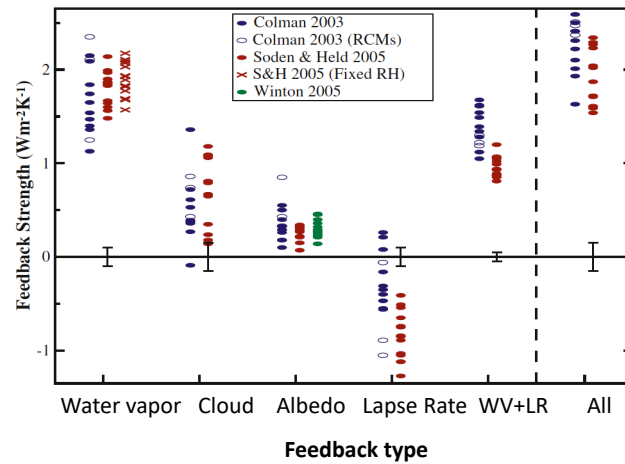


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8

8

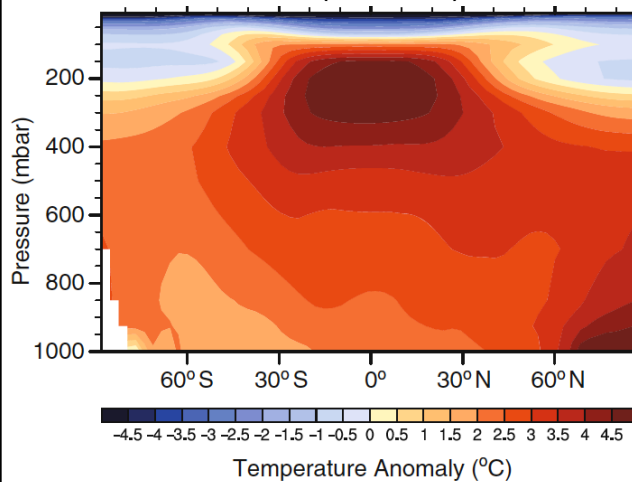
Summary regarding feedbacks



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9

2080-2099 minus 1980-2999, A1B scenarios,
mean of 21 GCMs (IPCC, 2007)



Q. Which feedback?

- Water vapor feedback
- Ice-albedo feedback
- Lapse-rate feedback

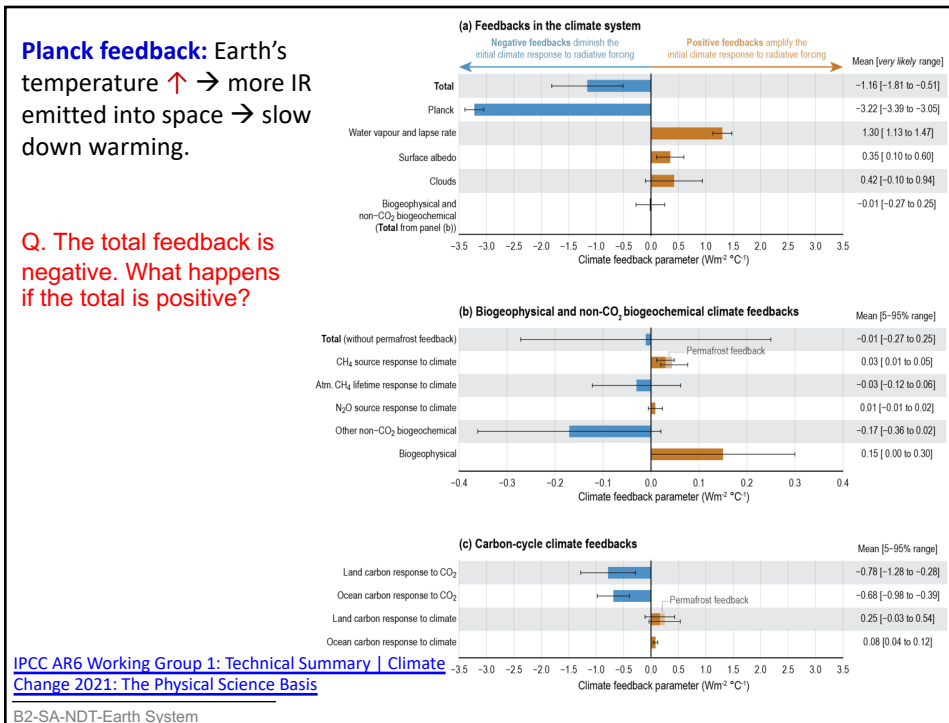
Q. Why is the stratosphere cooling?

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10

Planck feedback: Earth's temperature $\uparrow \rightarrow$ more IR emitted into space $\rightarrow$ slow down warming.

Q. The total feedback is negative. What happens if the total is positive?



11

Runaway greenhouse effect

- A runaway feedback occurs when the **positive feedback** (amplifiers) becomes stronger than the **negative feedback** (stabilizers)
- Results:** The planet enters a state of continuous warming that cannot be stopped by internal natural processes
- Key driver:** The failure of the **Planck feedback**.
In a runaway scenario, the atmosphere becomes so thick with greenhouse gases (e.g., water vapor, CO₂, etc.) that it no longer allows more heat to escape into space.

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12

12

Runaway greenhouse effect: Earth vs. Venus

	Earth (Stable)	Venus (Runaway)
Total Feedback	Negative (Stabilizing)	Positive (Unstable)
State of Water	Liquid/Solid/Gas	Vapor only (lost to space)
Surface Temp	~15°C	~460°C
Atmosphere	99% Nitrogen/Oxygen	96% CO ₂

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13

13

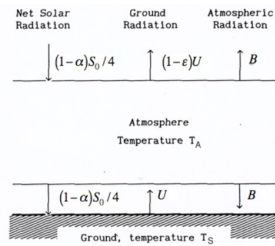
How to quantify the effects of feedback processes?

- Using models

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14

Example: (from the previous lesson)



$$T_S = \left(\frac{2}{2-\epsilon} \right)^{1/4} T_E$$

The **strength of greenhouse effect** is measured through the so-called **back radiation**:

$$F_{\downarrow} = B = \epsilon \sigma T_A^4 = \frac{\epsilon}{2-\epsilon} \sigma T_E^4 \equiv \frac{\epsilon}{2-\epsilon} Q.$$

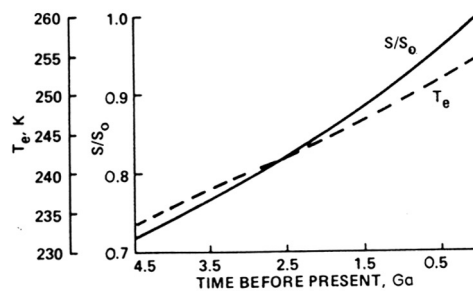
- $\epsilon \uparrow \rightarrow B \uparrow \rightarrow T_S \uparrow$
- Q. Planck feedback?
- Q. Water vapor feedback?

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15

Question

- Why has the temp of the Earth's surface remained in a narrow range for the last 3.6 billion years when the heat of sun has increased by 25%?



EBM-0D $\rightarrow$? Temperature

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16

16

Faint Sun paradox

Solar Irradiance increases by 25 to 30% over 4.5 Gy. Forcing: 100 Wm^{-2}

What did happen?	What should have happened?
Water (liquid) and life existed ~ 4Ga	Water frozen for ~2 Gy
Low-latitude glaciation 750, 600, and 300 Ma	Earth → out of ice age conditions by 700 Ma
Earth appears to have cooled over past 60 My	Warm up of Earth over time

- What else happened?**
- GHG concentration
 - Tectonics
 - Milankovitch cycles
 - etc.

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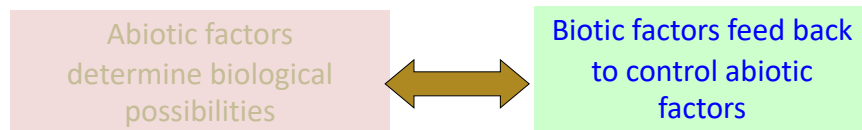
17

17

Gaia Hypothesis

James Lovelock, 1970s

Biota co-evolves with their environment: **biota might optimize their abiotic environment by means of negative feedback**



e.g. T increase → sparser vegetation, more desertification → increase albedo → reduce temperature (a **negative feedback loop**)

Earth can't be understood without considering the role of life

→ Daisy World model: a simple biospheric feedback model

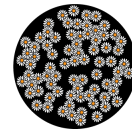
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18

18

Daisy world

- Daisyworld is an imaginary planet: transparent atmosphere, free from clouds & greenhouse gases
- Flat planet → similar changes in temperature with changing solar luminosity; does not experience any seasonality in climate
- The planet's biota:
 - Black daisy → low albedo → absorb more energy → warming
 - White daisy → high albedo → reflect more energy → cooling
- Conditions on Daisy World are suitable for the growth of daisies over the entire surface of the planet

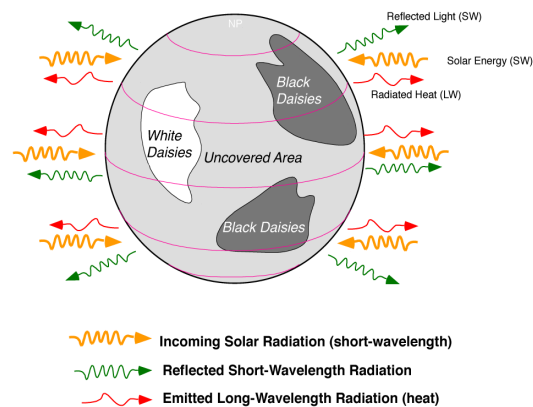


19

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19

Daisy World – two species



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20

Assumptions

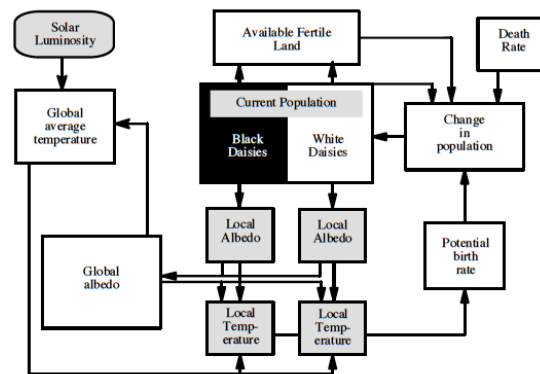
1. The rate of population change $\rightarrow$ depends on the death rate and the potential birth rate for that species, and the amount of fertile land available for growth
2. The birth rate for both species of daisy depends on the local temperature
3. The local temperature depends on the difference between the global and local albedo, and on the global temperature
4. The global temperature depends on the luminosity of the Sun and the planetary albedo
5. The planetary albedo is the sum of the local albedo components
6. The amount of fertile land available for further growth of the daisies depends on the total amount of land and the current coverage of the two species of daisy

21

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21

Daisyworld Model: Graphical representation



22

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22

Mathematical formulation

1. The amount of fertile land available for daisy growth:

$$x=1-w-b \quad (1)$$

where: x is the amount of available land; w, b are areas of white & black daisies, respectively

2. The planetary albedo:

$$\alpha_p = w\alpha_w + b\alpha_b + x\alpha_g \quad (2)$$

where: $\alpha_w, \alpha_b, \alpha_g$ are the albedos of white daisy, black daisy & bare ground respectively

3. The planetary temperature: $T_e = \sqrt[4]{\frac{S(1-\alpha_p)}{4\varepsilon\sigma}} - 273.15 \quad (3)$

23

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23

Mathematical formulation

4. The **local temperature** for populations of black & white daisies:

$$T_b = T_e + q(\alpha_p - \alpha_b) \quad (4)$$

$$T_w = T_e + q(\alpha_p - \alpha_w) \quad (5)$$

where: q is the constant used to calculate local temperature as a function of albedo

24

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24

Mathematical formulation

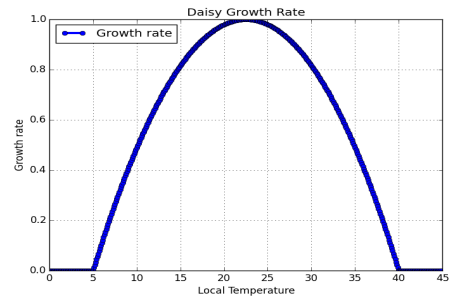
5. The **growth rate** of the populations of black and white daisies:

$$f_b = \max(0, 1 - k (T_0 - T_b)^2) \quad (6)$$

$$f_w = \max(0, 1 - k (T_0 - T_w)^2) \quad (7)$$

where: k and T_0 are constants so that growth occurs within a range of temperature and peaks at T_0 .

$k=0.003265$; $T_0=22.5 \rightarrow$ growth occurs between 5-40°C, peaks at 22.5°C



25

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25

Mathematical formulation

6. The change in area of black and white daisies over time:

$$db/dt = b(xf_b - d) \quad (8)$$

$$dw/dt = w(xf_w - d) \quad (9)$$

where: d is the death rate; t is time

7. The new area of black and white daisies :

$$b = b + db/dt \quad (10)$$

$$w = w + dw/dt \quad (11)$$

26

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26

Model parameters

- S : 1000 W/m²
- $\varepsilon=0.3$: emissivity of the atmosphere
- $\sigma=5.67 \times 10^{-8}$
- $\alpha_w=0.75$, $\alpha_b=0.25$, $\alpha_g=0.5$
- $q=20$
- $k=0.003265$
- $T_0=22.5^\circ\text{C}$
- $d=0.1$
- At the initial stage $\rightarrow b=w=0.2$

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27

27

Model outputs

- T_e : $^\circ\text{C}$, planetary temperature
- α_p : unitless, planetary albedo
- T_w , T_b : $^\circ\text{C}$, local temperature of white daisies and black daisies
- f_w, f_b : (inverse time) growth rate of white daisies & black daisies
- w, b, x : fraction of the planet's surface covered by white daisies, black daisies and bare ground

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28

28

Algorithm

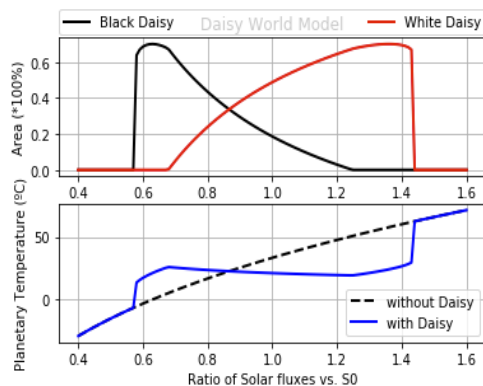
- The algorithm for Daisy World follows the logic of the model equations (1) to (11) above.
- After Eq(11), need to return to step 1, i.e. Eq(1) and repeat until the system reaches **equilibrium** (i.e. model's variables are almost unchanged)

29

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29

Results



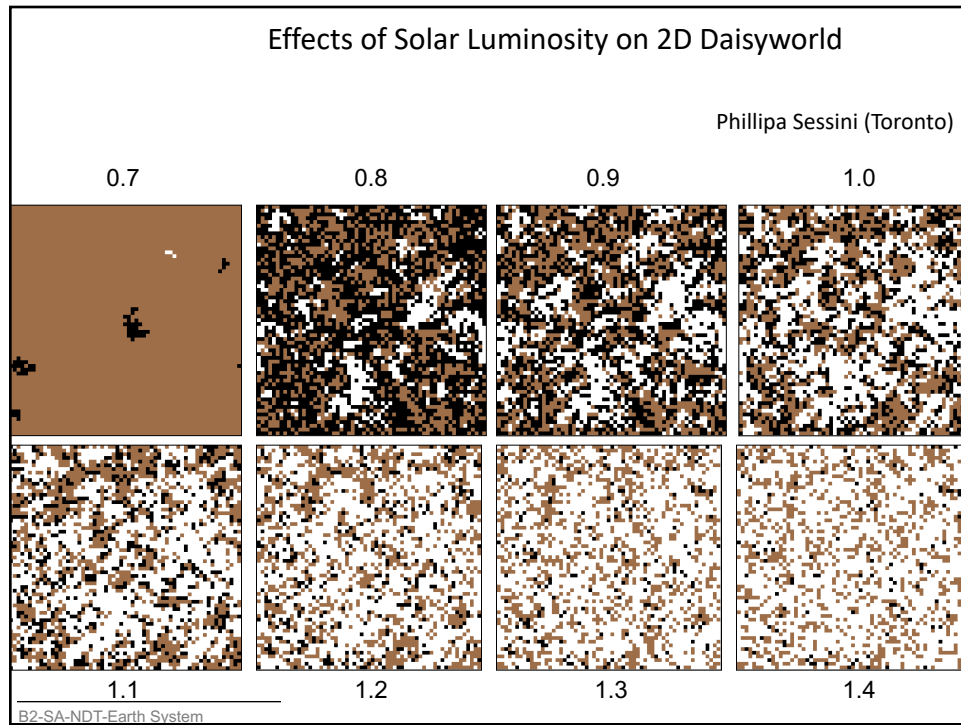
- When planet is **cold** :
 - **white daisies** are cooler than surrounds → hinder their own growth
 - **black ones** absorb the sunlight → warm up their surrounds → & warm up the whole planet.
- When planet is **hot**, the reverse happens, and **white daisies** dominate.

- The daisies, only with their albedo and birthrate, maintain the living planet's temperature at a daisy-friendly 20–30°C when the solar input varies

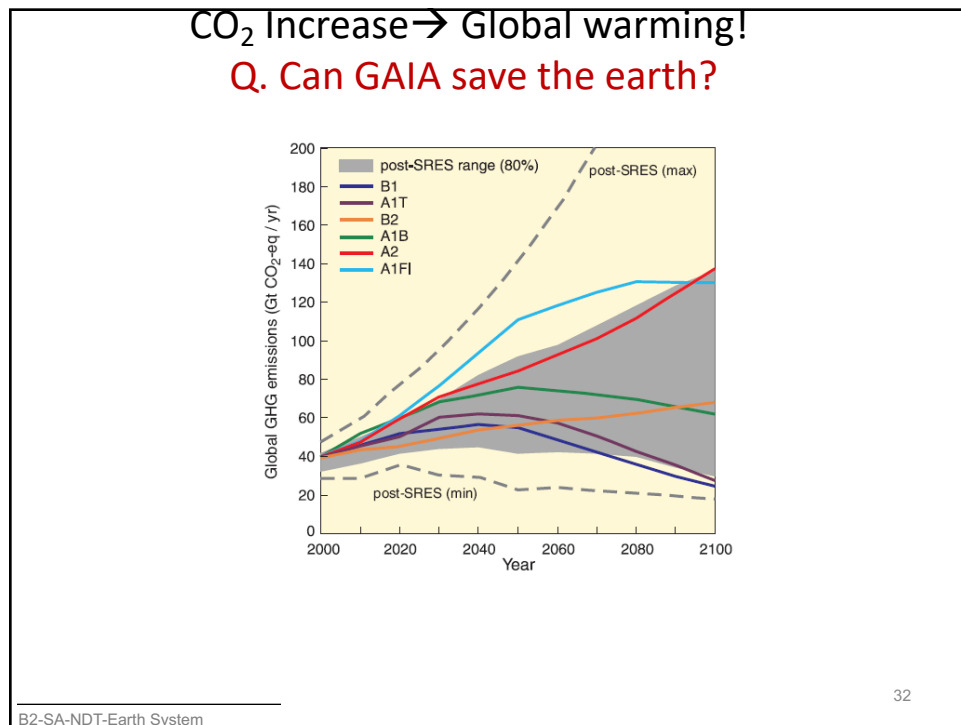
30

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30



31



32

32

Practice#6: Daisy World

Write a program to represent the Daisy World with two species: black and white.

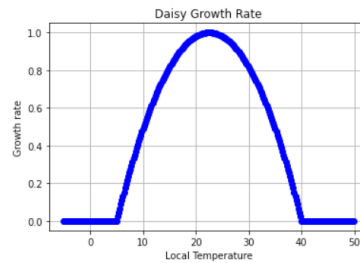
1. Plot the growth rate function f_b & f_w versus temperature

$$f_b = \max(0, 1 - k(T_0 - T_b)^2)$$

$$f_w = \max(0, 1 - k(T_0 - T_w)^2)$$

$$k = 0.003265$$

$$T_0 = 22.5^\circ\text{C}$$

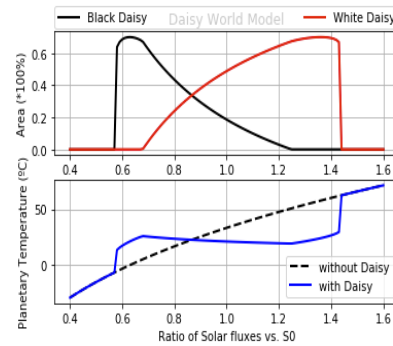


2. $S = S_0 = 1000$; at the initial stage $b = w = 0.2$. Find the equilibrium b and w values

3. S/S_0 varies from 0.4 to 1.6

a) Plot the equilibrium daisy black/white area versus S/S_0 .

b) Plot the planetary temperature versus S/S_0 .



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